CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

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FRUIT AND ITS COLOR USING BACKTRACKING

Aim:

To implement backtracking in Prolog to identify fruits and their colors.

Objective:

To demonstrate how Prolog searches through multiple possible solutions using backtracking.

Algorithm:

1. **Start.**
2. **Define fruit-color facts.**
3. **Query a fruit or color.**
4. **Prolog searches for a matching fact.**
5. **If multiple matches exist, Prolog uses backtracking to find the next solution.**
6. **Display all possible solutions.**
7. **Stop.**

Flowchart:

```
START
  |
  v
Store Fruit-Color Facts
  |
  v
Enter Fruit/Color
  |
  v
Search Matching Fact
  |
  v
Match Found?
 /       \
Yes       No
 |         |
Display    Fail
Solution
 |
 v
More Solutions?
 /       \
Yes       No
```

Backtrack STOP

Prolog Program:

% Fruit and Color using Backtracking

fruit_color(apple, red).

fruit_color(apple, green).

fruit_color(banana, yellow).

fruit_color(orange, orange).

fruit_color(grapes, green).

fruit_color(grapes, purple).

fruit_color(mango, yellow).

fruit_color(strawberry, red).

Query

?- fruit_color(apple, Color).

Output:

Color = red ;

Color = green.

Another query:

?- fruit_color(Fruit, green).

Output:

```
93 ?- fruit_by_color(yellow, Fruit).
Fruit = banana
```

Result:

Thus, fruit and color information is successfully retrieved using Prolog backtracking.